

CLASSICAL VERSUS QUANTUM COMPUTING

Feature	Classical Computing	Quantum Computing
Basic unit of information	Bit (0 or 1)	Qubit (can be 0, 1, or both at the same time due to superposition)
State representation	Binary states (either 0 or 1)	Superposition (0 and 1 simultaneously)
Processing method	Sequential, one operation at a time	Parallel, multiple calculations simultaneously
Computation speed	Limited by processing power, slower for complex problems	Potentially exponentially faster for certain problems (e.g., factoring large numbers)
Hardware	Based on traditional semiconductor technology (transistors)	Requires specialised quantum hardware (e.g., superconducting qubits, ion traps)
Current use	Used for almost all everyday applications	Experimental, used in specialised fields like research, finance, and healthcare
Availability	Widely available, accessible to everyone	Still in development, limited access (cloud-based quantum computers are becoming more common)
Scalability	Easily scalable with improvements in hardware	Challenges in scaling require breakthroughs in quantum technology

problems much faster and more efficiently than traditional systems.

Main challenges of quantum computing adoption

High costs of quantum hardware.

Quantum systems require expensive, specialised components (e.g., superconducting qubits, cryogenic cooling), limiting access to small organisations and research labs.

Technical complexity and steep learning curve. Quantum computing involves complex principles (e.g., superposition, entanglement) that demand specialised knowledge, creating a steep learning curve for developers.

Limited access to quantum resources. Quantum hardware is scarce, and cloud platforms offer limited computational power and user access, slowing down development and innovation.

Quantum error correction and stability. Quantum computers are sensitive to noise and interference, making it challenging to maintain accuracy. Error correction methods are still evolving and not fully reliable.

The role of open source in quantum computing

In the context of quantum computing, open source platforms allow developers, researchers, and organisations to collaborate, share resources, and build on existing

tools without the need for costly proprietary solutions. Some key benefits of open source in quantum technology include:

Collaboration. Developers and researchers can work together to accelerate innovation.

Transparency. The open nature of the code allows for deeper understanding and trust in the software.

Cost-effective. Open source platforms eliminate licensing fees, making quantum computing more accessible to a wider range of users.

Flexibility. Users can modify the code to meet their specific needs, fostering creativity and customisation.

Open source platforms play a crucial role in democratising quantum computing by lowering technical and financial barriers. With open source tools, developers and researchers, especially those from smaller institutions or startups, can access and experiment with quantum computing software without needing to invest heavily in proprietary technology. This accessibility fosters wider participation and speeds up the development of quantum algorithms and applications.

NOTABLE OPEN SOURCE QUANTUM COMPUTING PROJECTS

Several notable open source quantum computing projects have gained

attention for their contributions to the field. These projects provide tools, frameworks, and platforms for researchers and developers to explore quantum algorithms and technologies. Some of the most prominent ones are discussed briefly below.

IBM's Qiskit

Qiskit is an open source quantum computing software development kit (SDK) created by IBM. It allows users to write quantum algorithms, simulate quantum circuits, and run them on IBM's quantum computers. Its main features are:

Quantum circuit creation.

Allows users to create quantum circuits with ease, providing a high-level interface for building quantum programs.

Simulation. It supports the simulation of quantum circuits on classical computers, enabling algorithm development before execution on real quantum hardware.

Real hardware integration.

Provides access to IBM's quantum computers via the IBM Quantum Experience platform, allowing users to run algorithms on actual quantum processors.

Cross-platform compatibility.

Can run on a range of devices, from personal computers to large-scale quantum processors.

Extensive libraries. Includes a variety of libraries for quantum machine learning, optimisation, and